Fall 2015 / EC571

Instructor:Ajay Joshi

Exam 2

Date: November 5, 2015

Name : _____

BU ID : _____

Question	Maximum points	Points scored
1	20	
2	20	
3	20	
4	20	
5	20	
6	20	
Bonus	5	
Total	100	

1. The duration of the exam is 2 hours.
2. The maximum grade is 100 points.
3. Please write your name on all pages.
4. Attempt any 5 of the 6 questions. Please indicate the question number that should not be graded.
5. The solutions should be written in the spaces provided on the test sheet.
6. If you need to use extra sheets of paper, write your name and BU ID on it, and clearly indicate the question number.
7. Material sharing is not allowed.
8. Calculators are allowed.
9. Boston University Academic Code of Conduct applies.

Question 1

Sketch the schematics for $f(P,Q,R,S) = \overline{P \cdot (Q+R) + S}$ using each of the following circuit techniques. You can assume that you have both complemented and non-complemented inputs:

1. Static CMOS. (5 points)

2. Pseudo-nMOS. (5 points)

3. Dual-rail domino. (5 points)

4. Pass transistors (including the inverter at the end). (5 points)

Question 2

Consider the circuit shown below. Assume a unit clk-to-q delay for the register, two units of delay for the multiplexer and two units of delay for each logic block. The registers are positive-edge triggered and have a setup time of one unit.

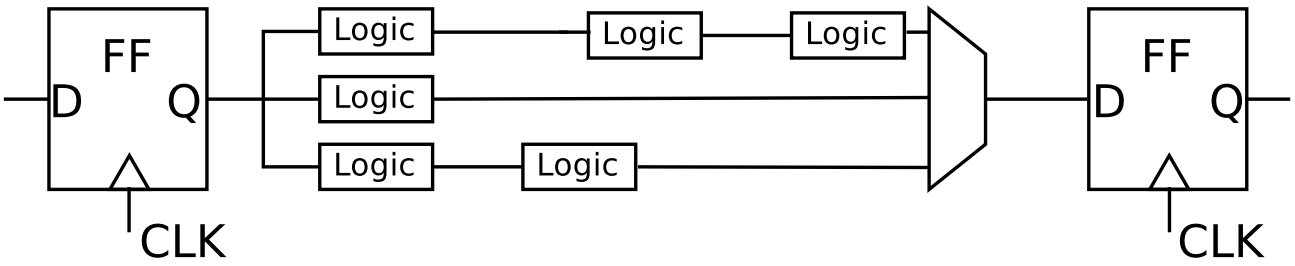


Figure 1

1. Determine the minimum clock period for the above circuit. (7 points)

2. Assuming a hold time of 2 time unit, what is the maximum positive clock skew that can be tolerated by this circuit? (7 points)

3. If the clock input at both FFs has a jitter of ± 1 unit, what would be the minimum clock period of the circuit? Assume zero clock skew (8 points)

Question 3

Compute the probability of the energy consuming transitions of the output for logic function $f_1 = \overline{A \cdot B + C}$ implemented in static complementary CMOS logic and dynamic logic. The probability of having 0 and 1 as the input at A, B and C is as follows: P(A=0) = 0.3, P(A=1) = 0.7, P(B=0) = 0.4, P(B=1) = 0.6, P(C=0) = 0.1 and P(C=1) = 0.9. *Hint: You need to determining the probability of 0 to 1 transition for both logic style implementations of the function. (20 points)*

Question 4

For the circuit shown in Figure 2, sketch the timing waveforms of the voltages at *clk, A, B, C, D, out, X, Y & Z* for the given inputs (Table 1). Also calculate the energy drawn from the power supply during each half of the clock cycles over the 5 clock cycles listed in Table 1. Here τ_{clk} is the clock period. Assume that the voltage at clk, A, B, C and D changes with a ramp at the time stamps given in the table. You can approximate the time scale, but be sure to compute the voltage levels. Assume V_{DD} = 2.5 V and V_T = 0.5 V. Also assume, that the clock period is long enough for discharging the four capacitors. (20 points)

Time	Clock (ϕ)	A	B	C	D
0	0	0	0	1	0
$\tau_{clk}/2$	1	0	0	1	0
τ_{clk}	0	0	1	0	1
$3\tau_{clk}/2$	1	0	1	0	1
$2\tau_{clk}$	0	1	1	1	0
$5\tau_{clk}/2$	1	1	1	1	0
$3\tau_{clk}$	0	1	0	1	0
$7\tau_{clk}/2$	1	1	0	1	0
$4\tau_{clk}$	0	1	1	0	1
$9\tau_{clk}/2$	1	1	1	0	1
$5\tau_{clk}$	0	1	1	0	0

Table 1

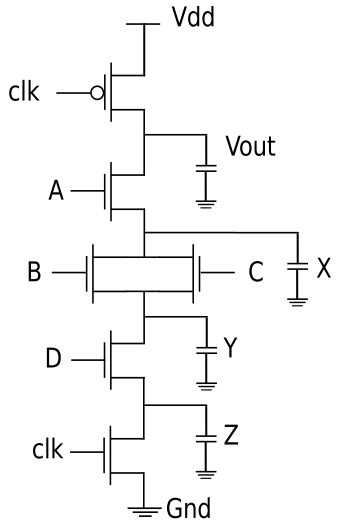


Figure 2

Question 5

1. Implement the following function f_1 using NMOS-based pass-transistor logic: $f_1(A,B,C) = \overline{(A+B)\cdot C}$. Include an inverter at the end to achieve rail-to-rail output. Ignore the keeper transistors. (7 points)

2. Draw the RC schematic for this circuit Assume your circuit has to drive a load capacitor C_L. (6 points)

3. Assuming all NMOS transistors in the pass-transistor component are sized the same, which set of inputs result in the largest delay? Write the Elmore delay equation for that case. *Hint: The total delay will be the sum of the delay of the pass-transistor component + the inverter.* (7 points)

Question 6

1. Figure 3 shows the schematic of a Switched Output Differential Structure (SODS) logic. Explain how the logic style operates. Comment on the strengths and weaknesses of this design style. You can assume that both NAND and AND have 2 inputs and you have access to A, B, A' and B' (10 points).

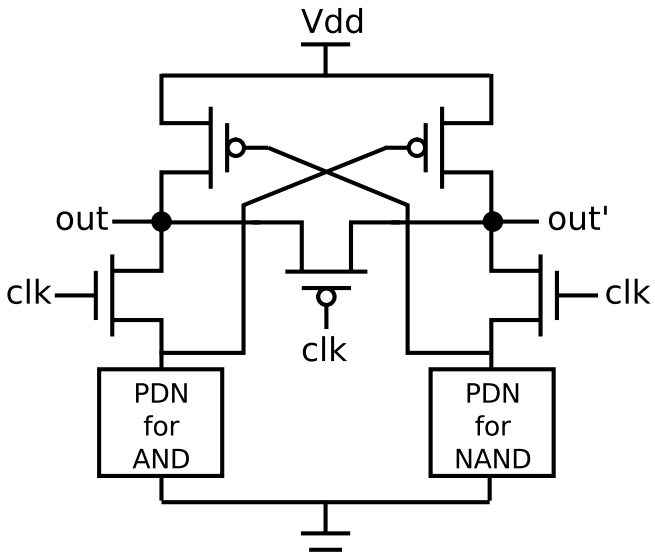


Figure 3

2. Design a Master-slave negative edge-triggered register using latches. You can draw multiplexer-based implementation for your latches. Define setup time, propagation delay (clock-to-Q delay) and hold time. (10 points)

Bonus question (5 points)
Prof. Joshi is not much into winter sports, but is considering trying 2-3 different winter sports this year. Please suggest five different winter sports that he should try.